

Dwell-Adaptive RF Signal Classification from Duration-Complete Synthetic Data

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Abstract

Three facts about burst-structured radio are uncontroversial: the identity of protocols such as TDMA cellular and frequency-hopped Bluetooth lives in timing structure at millisecond scale; such emitters are silent most of the time; and the common practice of capturing a fixed number of samples across swept sample rates yields training windows that span microseconds. The conclusion is immediate—a classifier cannot learn structure its training windows do not contain, and rejecting “empty” windows discards the one statistic that separates low-duty-cycle classes—yet its consequence is routinely misattributed to model capacity. We propose treating *observation time* as a *first-class variable* on both sides of learning: DACS (Dwell-Adaptive Classification from duration-complete Synthesis) trains a multi-objective network—prototype classification, masked denoising against clean references, a worst-case-over-dwell loss, and a learned confidence signal—on captures whose durations contain each protocol’s defining timescales and whose silences appear at their true class-conditional rates; at inference the confidence signal chooses the dwell, answering early when sure and escalating when not. With architecture, optimizer, and budget held fixed, duration completeness alone lifts 7-class balanced accuracy from 0.503 (shortest fixed-count window tier) to 0.954 at 1 ms dwell and 1.000 at 10 ms, replicated on two seeds; architecture substitution, longer training, and worst-case reweighting on fixed-count data all fail to approach this effect. A companion analysis shows the field’s worst-cell metrics are order statistics that cannot display such gains at customary cell sizes, and prescribes cell designs that can. Training cost is linear in dwell.

Index terms—RF signal classification, selective classification, anytime inference, synthetic training data, frequency hopping.

1 Introduction

Start from premises no RF engineer disputes. *Premise 1*: the identity of burst-structured protocols is temporal. A GSM carrier transmits one 577 μ s burst per 4.615 ms TDMA frame [14]; Bluetooth Classic occupies one of 79 channels for one 625 μ s slot before hopping; BLE advertising transmits three sub-160 μ s packets every few tens of milliseconds [13]. What distinguishes these classes from one another—and from continuous carriers—is *when* energy appears, not only what it looks like. *Premise 2*: such emitters are silent most of the time; si-

lence is not an aberration of the capture but the dominant state of the signal. *Premise 3*: prevailing practice fixes capture length in *samples* while sweeping sample rate, so at high rates a training window spans microseconds of physical time.

The conclusion follows without any appeal to learning theory: a window shorter than a protocol’s defining period cannot contain its identity, so a classifier trained on such windows faces “a narrowband blip or nothing” (Fig. 1) regardless of architecture—and a pipeline that rejects near-empty windows as invalid additionally deletes $P(\text{silence} \mid \text{class})$, the one signature premises 1 and 2 jointly guarantee. Sec. 2 verifies each premise quantitatively and adds a third, less obvious casualty: the minimum-over-cells metrics used to accept such systems are order statistics whose readout sits far below true per-cell recall, so even a solved problem cannot *display* as solved.

This diagnosis invites a simple idea: make *observation time* a first-class variable. On the training side, captures should span durations commensurate with each protocol’s timescales, with silence retained at its true class-conditional rate—a property of the data we call *duration completeness*. On the inference side, once models are trained across dwells, the dwell itself becomes a decision: classify at the shortest dwell, and let a learned confidence signal decide whether to answer or to keep listening. DACS (Dwell-Adaptive Classification from duration-complete Synthesis) realizes both halves (Secs. 3–4), and controlled experiments (Sec. 5) verify the causal claim: with architecture, optimizer, and budget fixed, duration completeness alone moves 7-class balanced accuracy from 0.503 to 0.954 at 1 ms and to 1.000 at 10 ms on both tested seeds, while architecture substitution, tripled training budget, and worst-case reweighting on fixed-count data fail to approach it.

Relation to prior work. Deep I/Q classification is established [1, 2]; our contribution is orthogonal to architecture. The model composes known elements—prototype heads [3], masked denoising [4, 5], trained confidence [6, 7]—but the dwell policy instantiates selective prediction [8] and early exiting [10, 11] along a new axis: physical observation time rather than network depth, with calibration [9] governing escalation.

2 The Premises, Measured

We quantify each premise on the *fixed-count control corpus*: captures of 16,384 samples at rates swept over 1–200 Msps (median TDMA-class window 1.55 ms), constructed to instan-

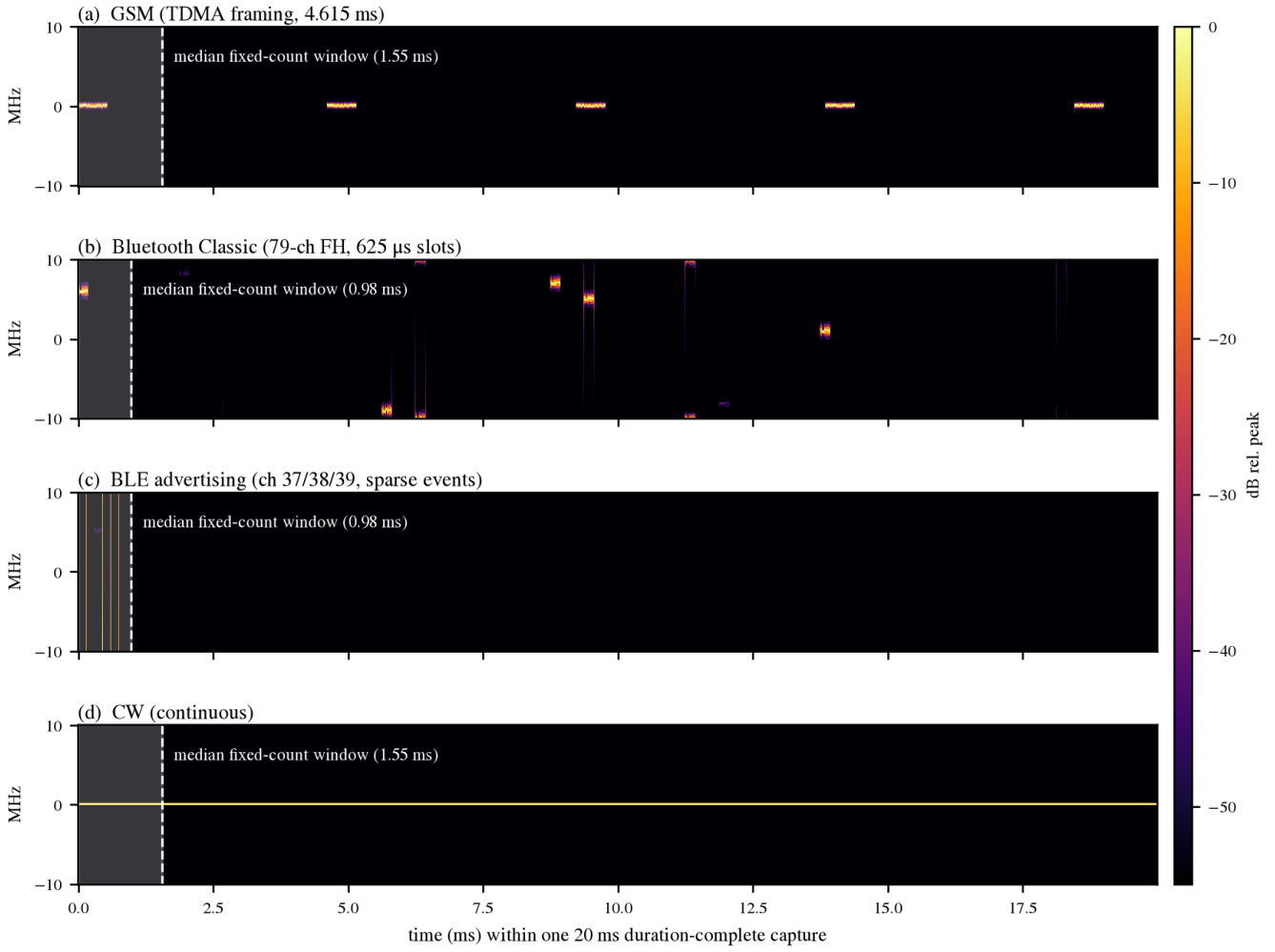


Figure 1: Class identity lives at timescales a fixed-sample-count window rarely contains. Spectrograms of single 20 ms duration-complete captures (clean synthesis, 20 Msps): (a) GSM, (b) Bluetooth Classic, (c) BLE advertising, (d) CW. Shading marks the median window obtained by fixing capture length in samples while sweeping sample rate (here 16,384 samples over 1–200 Msps). Within that window, GSM shows at most one burst, Bluetooth Classic at most one hop, and BLE one event or pure silence. The 20 ms capture contains the discriminants: TDMA framing [14], slot-clocked hopping and sparse advertising events [13], continuity for CW.

tiate the prevailing convention with the same classes and impairment envelope as our method’s corpus. We call such data *duration-starved*. Throughout, *worst-window balanced accuracy* denotes the minimum over the corpus’s three window tiers (4,096/8,192/16,384 samples) of 7-class balanced accuracy.

Silence dominates (premises 1–2). Against the stationary expectation that the first quarter of a capture holds 25% of its energy, 37% of GSM captures hold <10% of capture energy in their first 4,096 samples (Fig. 2a aggregates all GSM configurations; the burst-mode configurations individually range 40–48%, while the always-on broadcast carrier is $\approx 0\%$; block crest factor 39 $\times$); frequency-hopped captures 23%; DSSS 24%; continuous classes $\approx 0\%$. Conventional pipelines *reject* empty windows as invalid, discarding exactly the statistic that separates low-duty-cycle classes. A prior-optimal treatment of empty windows alone bounds balanced accuracy near 0.93.

Short windows erase identity (premises 1 and 3). The continuously transmitting GSM configuration (loaded broadcast

carrier)—never empty—still yields only 0.54 recall on the shortest window tier under the feature-engineered baseline of Sec. 5 (mean over its 24 tuned configurations): the GSM/Bluetooth discriminant is frame timing versus hop timing, both invisible inside a sub-millisecond span. Energy presence is not the limiting factor; temporal structure is.

The metric cannot display success. Acceptance criteria for deployed classifiers often take the form of minimum recall over (configuration $\times$ condition) cells. With J cells of n evaluation rows, the displayed value is $\mathbb{E}[\min_j \hat{p}_j]$ with $\hat{p}_j \sim \text{Bin}(n, p)/n$; for a representative design with $J=124$ cells of $n=16$ (e.g., 31 configurations crossed with 4 conditions), true per-cell recall 0.95/0.97/0.99 reads as 0.77/0.82/0.89 (Fig. 2b). A system can be near-ceiling while the metric reports 0.8. The remedy is not a different statistic but adequate cells: at $J=34$, $n=96$, a true 0.97 displays as 0.93, and ≥ 0.9 scores become achievable in principle rather than statistically excluded. We adopt per-configuration cells of $n=96$ per dwell throughout, reporting bal-

anced accuracy alongside the minimum cell.

3 The DACS Classifier

The idea’s inference half is a network that treats dwell as a decision variable. A single spectro-temporal encoder (STFT front end; six 3×3 convolution–GroupNorm–SiLU blocks, fully convolutional in time, $32 \times$ temporal downsampling; 1.0M parameters alone, 1.57M with the decoder and heads below) feeds three heads (Fig. 3). With episode windows x at dwells drawn from an ordered set $\ell \in \{\ell_1 < \dots < \ell_K\}$ (we use $\{1, 2.5, 10\}$ ms), spectrogram S_x , random time-frame mask m , and class prototypes computed from support embeddings [3]:

$$\mathcal{L} = \mathcal{L}_{\text{proto}} + \lambda_r \|D(E(m \odot S_x)) - S_{\text{clean}}\|_1 \\ + \lambda_d(t) \mathbb{E}_t \left[\max_{\ell} \mathcal{L}_{\text{proto}}(x_i^{(\ell)}) \right] + \lambda_c \text{BCE}(g(x), \mathbb{I}[\hat{y} = y]),$$

where $\mathcal{L}_{\text{proto}}$ is episodic cross-entropy over squared prototype distances with a learned temperature, $D \circ E$ is a masked denoising path supervised by the capture’s clean synthesis reference [4, 5]—silence is thereby a reconstruction target, not an exclusion—the third term trains the shortest dwell against its hardest cases (prototypes detached; weight ramped linearly from zero), and g is a confidence head trained on realized correctness [6, 7]. Optimization uses AdamW with cosine annealing [12].

Dwell policy. At inference the system classifies at ℓ_1 and consults g : if calibrated confidence [9] clears a threshold the answer is emitted; otherwise the capture extends to the next dwell—selective classification [8] in which the cost of abstention is observation time, and the analogue of early exiting [10, 11] with physical time replacing depth.

4 Training Data

The idea’s training half requires corpora whose captures contain the structure of Fig. 1; we summarize the generation technique and refer implementation details to the artifact. Every waveform generator is *index-pure*—a deterministic function $y[n] = G(n)$ of the absolute sample index—so captures of arbitrary duration assemble byte-exactly from bounded, individually validated packet content. Aperiodic burst processes are composed by replaying standards-derived packet waveforms [13] on unbounded timelines: slot-clocked channel hopping with keyed-hash channel selection and utilization for Bluetooth Classic, and an event grid with the specification’s pseudo-random *advDelay* for BLE advertising; the compositions reproduce occupancy and timing *statistics*, with the specification’s hop-selection kernel and payload variation explicitly out of scope. Clean captures are synthesized at native rates, resampled to a common 20 Msps, and passed through a seeded receiver impairment chain (AWGN at $U(-5, 45)$ dB SNR referenced to active-sample power, LO-scaled carrier offset, phase noise, IQ imbalance, DC offset, static multipath), emitting row-aligned clean/impaired pairs; the clean side supervises the reconstruction head. Training windows are drawn by duration at random offsets, and empty windows are retained.

5 Experiments

Setup. Seven classes (am, bluetooth, cw, dsss, fm, gsm, ofdm) spanning 34 waveform configurations; 8,704 captures of 20 ms, 96 evaluation rows per configuration (160 further captures per configuration for training); impairments as in Sec. 4. The fixed-count control corpus is that of Sec. 2. Training budget denotes the number of episodic updates (7-way, 5-shot/5-query; dwell drawn uniformly per episode); the base budget is 8,000 episodes and “ $3 \times$ ” is 24,000. Optimization: AdamW [12], cosine annealing, peak learning rate 2×10^{-3} ; loss weights $\lambda_r=0.3$, $\lambda_d=0.15$ (linear ramp over the first tenth of training), $\lambda_c=0.1$. Evaluation classifies the capture-initial window of each evaluation row at each dwell, with class prototypes computed from 64 training captures per class under the identical protocol on both corpora.

Duration is the causal factor. Holding the encoder (1.0M parameters, prototype loss only—no reconstruction, worst-case, or confidence terms), optimizer, and budget fixed and swapping only the corpus (Fig. 4): balanced accuracy rises from 0.503 on the shortest fixed-count tier to 0.954 at 1 ms dwell, and from 0.682 to 1.000 at the longest tier vs. 10 ms, with every class including GSM and Bluetooth at recall 1.000 at 10 ms; an independent seed reproduces 0.958/0.972/1.000. Success criteria fixed before the runs were executed (GSM and BT ≥ 0.85 at 10 ms; short-dwell balanced $\geq$ the control level; monotone recall in dwell) all passed on both seeds. Because the control corpus sweeps sample rate, its tiers are matched to dwells by rank (shortest/mid/longest), not by physical duration, which fixed-count data cannot hold constant.

Nothing else closes the gap (Table 1). On the fixed-count control corpus: (i) the same spectro-temporal encoder *loses* to a feature-engineered baseline—hand-designed invariant patch preprocessing (sixteen normalized 64-sample patches with geometry canonicalization) feeding a compact CNN, tuned by a 50-trial hyperparameter search—scoring 0.34 (raw spectrogram input) and 0.45 (adding per-window RMS normalization and phase augmentation) vs. the baseline’s 0.49 GSM recall on the shortest tier: architecture is not the lever. (ii) Tripling the training budget *reduces* worst-window balanced accuracy, 0.681 $\rightarrow$ 0.657, the signature of shortcut overfitting on ambiguous windows. (iii) A worst-case-over-window auxiliary (max cross-entropy over window tiers) peaks at +0.009 over the 0.681 base-budget baseline (inverted-U dose response, weight ≈ 0.15) and recovers +0.055 at $3 \times$ budget (0.712 vs. 0.657): worst-case pressure helps but cannot supply missing information.

Full system. At an early checkpoint one third through its training schedule, the DACS network reaches balanced accuracy 0.909/0.950/0.990 at 1/2.5/10 ms with minimum-cell recall 0.448/0.760/0.948; surviving errors concentrate in single configurations at the shortest dwell—precisely the cases escalation targets. The raw confidence head predicts the answer-at-1 ms decision at 0.884 accuracy against a 0.909 marginal correctness rate at this checkpoint; the head is uncalibrated here, and we accordingly claim the escalation *mechanism*, deferring its operating curve to calibration on held-out data (Sec. 6). Training cost is linear in dwell: 1.4 ms (1 ms window) to 31 ms (20 ms window) per training step per window on one Apple

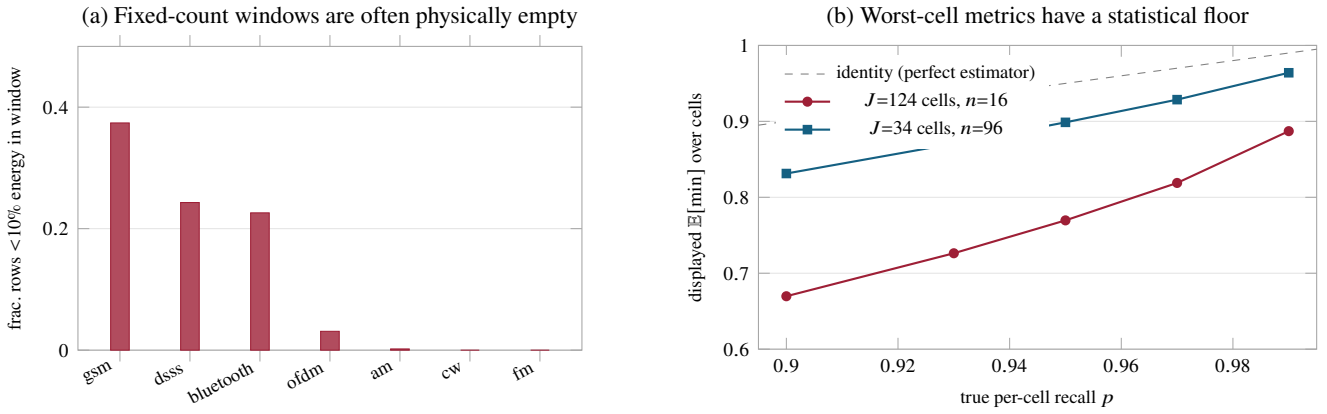


Figure 2: The premises, measured. (a) Fraction of fixed-count corpus rows whose scored window contains <10% of capture energy: the hardest classes are frequently *not in the window*. (b) Monte Carlo readout (2×10^4 draws) of minimum-over-cells metrics: small- n cells cap what any model can *display*, independent of its true performance.

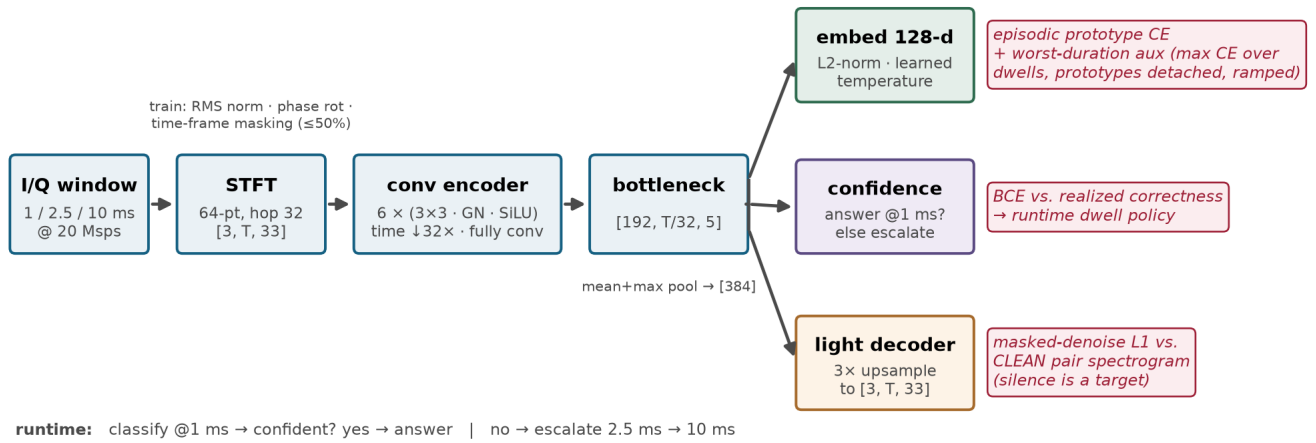


Figure 3: The DACS network. One shared spectro-temporal encoder; three heads. The prototype embedding [3] carries classification and preserves enrollment-based open-set operation; the confidence head [6, 8] drives the answer-early/escalate dwell policy; the light decoder reconstructs the clean reference’s spectrogram under time-frame masking [4, 5], making silence a learning target and regularizing against the shortcut overfit that duration-starved training exhibits (Sec. 5). The worst-case-over-dwell auxiliary trains the shortest dwell against its hardest cases.

Table 1: Controls on the fixed-count corpus: no model-side intervention approaches the effect of duration-complete data.

Intervention (control corpus)	Metric	Effect
Feature-eng. baseline, 50-trial search	worst-win. bal.	0.743
its GSM recall, shortest tier	GSM recall	0.49
Our encoder, raw / normalized input	GSM recall	0.34 / 0.45
3× training budget	worst-win. bal.	0.681 → 0.657
Worst-case loss, wt. .05/.15/.40	Δ w.-win.	+0.003/+0.009/−0.008
Worst-case loss, 3× budget	Δ w.-win.	+0.055
Duration-complete data, fixed encoder	shortest-tier bal.	0.503 → 0.954

M-series GPU, so duration completeness is computationally cheap.

6 Limitations

All data are synthetic; over-the-air validation remains future work [2]. The hop composition reproduces occupancy and timing statistics, not the specification’s selection kernel, and repeats verified payload bits. Captures contain a single emitter; co-channel interference and Doppler are absent. Confidence

calibration [9] on held-out data is required before escalation thresholds are meaningful, and the full-system numbers above are an early snapshot at one third of the training schedule.

7 Conclusion

Obvious premises—burst identity is temporal, silence is the dominant state, fixed-count windows span microseconds—force a simple conclusion: the binding constraint on burst-structured RF classification is what the training window can physically contain, and secondarily what the evaluation can statistically display. The idea that follows is to make observation time a first-class variable: DACS trains across dwells on duration-complete data with silence as evidence, and lets a learned confidence signal decide, capture by capture, how long to listen. The controlled result—0.503 to 0.954 at 1 ms with architecture fixed—and the failure of every model-side intervention to approach it suggest a general prescription: before scaling models on a plateaued worst-case metric, audit the observation window and the metric itself.

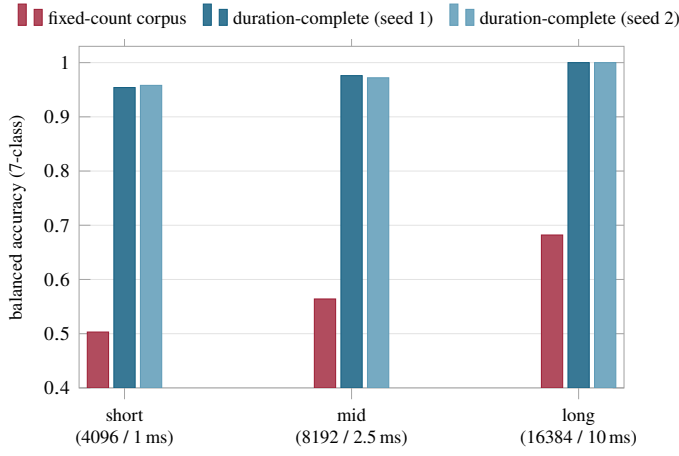


Figure 4: Controlled attribution. The identical 1.0M-parameter encoder, recipe, and budget on the fixed-count control corpus (window tiers in samples) vs. duration-complete data (dwells in milliseconds); tiers are matched by rank, not physical duration, which fixed-count data cannot hold constant. At 10 ms every class reaches recall 1.000 on both seeds.

References

- [1] T. J. O’Shea, J. Corgan, and T. C. Clancy, “Convolutional radio modulation recognition networks,” in *Proc. EANN*, 2016.
- [2] T. J. O’Shea, T. Roy, and T. C. Clancy, “Over-the-air deep learning based radio signal classification,” *IEEE J. Sel. Topics Signal Process.*, vol. 12, no. 1, 2018.
- [3] J. Snell, K. Swersky, and R. Zemel, “Prototypical networks for few-shot learning,” in *NeurIPS*, 2017.
- [4] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, “Masked autoencoders are scalable vision learners,” in *CVPR*, 2022.
- [5] P. Vincent, H. Larochelle, I. Lajoie, Y. Bengio, and P.-A. Manzagol, “Stacked denoising autoencoders: Learning useful representations in a deep network with a local denoising criterion,” *J. Mach. Learn. Res.*, vol. 11, 2010.
- [6] T. DeVries and G. W. Taylor, “Learning confidence for out-of-distribution detection in neural networks,” arXiv:1802.04865, 2018.
- [7] D. Hendrycks and K. Gimpel, “A baseline for detecting misclassified and out-of-distribution examples in neural networks,” in *ICLR*, 2017.
- [8] Y. Geifman and R. El-Yaniv, “Selective classification for deep neural networks,” in *NeurIPS*, 2017.
- [9] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *ICML*, 2017.
- [10] S. Teerapittayanon, B. McDanel, and H. T. Kung, “BranchyNet: Fast inference via early exiting from deep neural networks,” in *ICPR*, 2016.
- [11] G. Huang, D. Chen, T. Li, F. Wu, L. van der Maaten, and K. Q. Weinberger, “Multi-scale dense networks for resource efficient image classification,” in *ICLR*, 2018.
- [12] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *ICLR*, 2019.
- [13] Bluetooth SIG, *Bluetooth Core Specification*, Version 6.3, 2026.
- [14] 3GPP, “Multiplexing and multiple access on the radio path,” TS 45.002, Rel. 19, 2026.